

Appendix 1.1: SOULBOUND TOKEN (TRXS) SPECIFICATION

Executive-Level Technical and Regulatory Standard for Non-Transferable Governance and Competence Assets

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Compliance Framework: Swiss DLT-Act / ADGM / DIFC / W3C DID Standard / ISO-27001
Module Reference: MOD-CORE-TRXS-v3.4Architectural Principle: The Soulbound Token ($TRXS) serves as the immutable, non-transferable reputation and governance substrate of Transceptor Global. It bridges physical human contribution with cryptographic authority, completely decoupling financial capital from political power.

1. Architectural Invariants of TRXS

1.1. Absolute Non-Transferability and DID Anchoring

The $TRXS$ token is bound strictly to a single W3C Decentralized Identifier (`did_hash`) through cryptographic soulbinding. The token cannot be transferred, sold, leased, inherited, pledged as collateral, or assigned to third parties under any circumstances. Any attempt to initiate a peer-to-peer transfer or smart-contract delegation of $TRXS$ is rejected at the processor level within the Hardware Trusted Execution Environment (TEE).

1.2. Hardware TEE Isolation and Zeroization Protocol

All computation, quantization, and evaluation of $TRXS$ balances occur exclusively inside hardware-isolated execution environments (TEE / Open Enclave SDK / Strict PRM). Master decryption keys reside in volatile CPU cache. In the event of a physical boundary breach, memory tamper signal, or debug attempt, the core engine triggers an automatic zeroization routine (`SIGSEGV`), purging all key material within 10 milliseconds.

1.3. Commercialization and Monetization Ban

$TRXS$ is strictly separated from the asset and liquid value contour (T_{net} and fiat currencies). Direct or indirect purchase, sale, or financial conversion of $TRXS$ is defined as severe architectural fraud. Direct fiat input into the governance core is hard-blocked at the C++ kernel level (`clearing_gateway.hpp`). Discovery of commercialization schemes results in immediate DID revocation and permanent blacklisting.

2. Quantization and Electoral Weight Accumulation

2.1. Proof-of-Contribution Quantum Step

Electoral weight and competence status do not accrue continuously or passively; they increase in discrete quantum increments of $\Delta I_{\text{cp}} = +0.01$. Increments are granted strictly upon verified physical, cognitive, or scientific outcomes (Proof-of-Useful-Outcome / PoUCC). Raw credentials (ORCID, Scopus, GitHub Enterprise) serve only as baseline eligibility coefficients ($I_{\text{cp_raw}}$) and do not directly confer voting power without demonstrated real-world impact.

2.2. Dynamic Electoral Weight Corridor

The effective user voting weight (W_{user}) is algorithmically constrained within a strict meritocratic corridor:

$$W_{\text{user}} = \text{clip}\left(1.0 + \sum \Delta I_{\text{sc}} + \sum \Delta I_{\text{cp}} + \sum \Delta I_{\text{ft}}, , 0.01, , 5.00\right)$$

- **Baseline Status ($W_{\text{user}} = 1.00$):** Assigned upon standard identity verification (ZK-KYC) and baseline Liveness Check.
- **Technological Minimum ($W_{\text{user}} = 0.01$):** Applied as a non-zero floor for restricted, sanctioned, or degraded profiles.
- **Corridor Ceiling ($W_{\text{user}} = 5.00$):** Absolute maximum weight to prevent political monopoly or technocratic oligarchy.

2.3. Sliding Window Evaluation

Participant reliability and voting impact are evaluated over a rolling window of the last 50 verified decisions or RWA project milestones. Historical contributions outside the active 50-action window transition into passive baseline metrics, ensuring that governance authority reflects current competence rather than historical privilege.

3. Reputation Decay Model (SBT-Decay)

3.1. System Error Index (SEI) Calculation

To protect the system against incompetence, collusion, or decision drift, the kernel calculates a continuous System Error Index (SEI) over the 50-decision sliding window:

$$SEI = \frac{N_{\text{errors}} + N_{\text{deviations}}}{N_{\text{total_actions}}}$$

where N_{errors} represents votes cast in favor of failed initiatives ($I_{\text{sr}} < 1.0$) or substandard engineering audits.

3.2. Dual-Tier Decay Factor (α)

The decay multiplier (α) dynamically scales based on the nature of the error:

- **Base Expert Decay ($\alpha_{\text{base}} = 0.8$):** Applied to standard statistical noise, honest modeling divergence, or non-critical project variance.
- **Penalty Decay ($\alpha_{\text{penalty}} = 2.5$):** Triggered when a participant votes in favor of an initiative that exhibits critical financial fraud, gross environmental neglect, or security vulnerabilities.

3.3. Mathematical Formulation of Weight Adjustment

The user's effective weight adjustment factor (Λ_{user}) is calculated as an exponential decay function:

$$\Lambda_{\text{user}} = e^{-\alpha \cdot SEI}$$

$$W_{\text{effective}} = W_{\text{user}} \cdot \Lambda_{\text{user}}$$

When $SEI = 0$, $\Lambda_{\text{user}} = 1.0$, preserving full electoral weight. As errors accumulate, $W_{\text{effective}}$ decays smoothly toward the lower bound of 0.01.

4. Dynamic Preventive Compliance and Preemptive Oversight

4.1. Tranche Staging in RWA Projects

To prevent capital loss and governance failure, project execution and associated $TRXS$ reward distribution operate under a three-phase staging model:

1. **Phase 1 (Design & Approval):** Initial liquidity release against accumulated $TRXS$ reputation collateral; temporary advance index ($I_{\text{cp_advance}}$) granted.
2. **Phase 2 (Intermediate Milestones):** On-site verification via IoT sensors (LiDAR, GNSS, telemetry) unlocks secondary tranches.
3. **Phase 3 (Outcome Verification):** Final clearing after a mandatory operational period (90 to 365 days). Upon success, $I_{\text{cp_advance}}$ converts to permanent I_{cp}.

4.2. Pending Audit Status

If telemetry records a performance drop ($I_{\text{sr}} < 1.0$) or sensor anomaly, the project and the author's $TRXS$ profile enter `Pending Audit` status. Edge AI agents query adjacent nodes to calculate the Environmental Correlation Coefficient (K_{env}), separating physical force majeure from operational failure.

4.3. Preemptive Sanctions and Fraud Prevention

If the audit confirms deliberate fraud, bot manipulation, deepfake usage, or severe negligence prior to project completion, the system executes an automated preemptive sanction:

$$\Delta W_{\text{user}} = -4.99$$

This instantly collapses the author’s electoral weight to the minimum threshold ($W_{\text{user}} = 0.01$), freezes pending T_{net} tranches, and forfeits the initial guarantee stake to the recovery fund.

5. Time-Decay and Offline Hours

5.1. Domain-Specific Staleness Thresholds

Competence metrics are subject to domain-specific decay if a user remains inactive or fails to provide updated Proof-of-Outcome:

Domain	Staleness Threshold	Base Decay Factor (α)
IT, OpSec & Core Kernel	90 Days	1.20
RWA, Engineering & Energy	360 Days	0.80
Ecology & Regeneration	360 Days	0.50
Legal & RegTech Compliance	180 Days	0.90

5.2. Live Interactive Hours (H_{live}) Freeze Mechanism

The accumulation of expert weight requires verified interactive hours ($H_{\text{live}} \ge 100$ hours per sprint). If a participant goes offline or ceases active contribution, H_{live} enters a frozen state. During freezing:

- Existing $TRXS$ balances do not increase.
- Domain staleness timers continue to run, gradually applying decay until active re-verification occurs through a new Proof-of-Contribution sprint.